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Boron/Zinc Exchange Reaction in the Diastereoselective Arylation of *N*-Protected L-Prolinal

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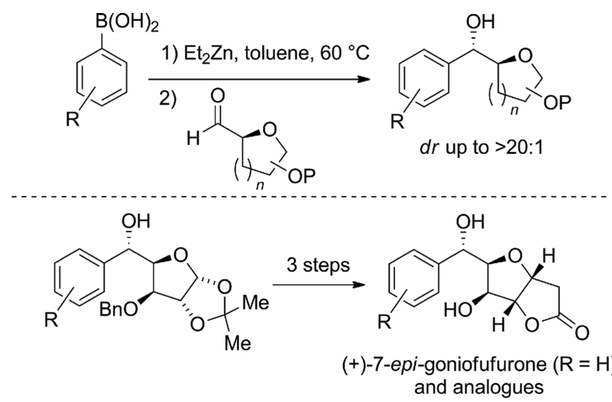
The diastereoselective arylation of chiral, non-racemic, *N*-trityl-protected L-prolinal has been investigated. The reactive aryl groups were generated by a boron/zinc exchange reaction between arylboronic acids and diethylzinc. The reactions proceeded in a highly diastereoselective fashion, and the resulting amino alcohols possessing two vicinal stereocenters were obtained in diastereomeric ratios of >20:1, regardless of the substituent at the transferable aryl group. The diastereoselectivity of the reactions is believed to be the result of an energetically favored Felkin–Anh transition state, with the competing Cram-chelation pathway precluded due to the presence of the bulky *N*-trityl protecting group.

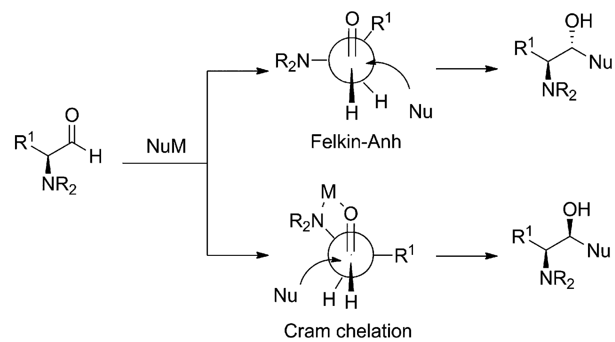
Introduction

Stereoselective reactions that take advantage of the existing stereochemistry in a given molecule to control the generation of new stereocenters are of great importance in organic chemistry, particularly in the synthesis of complex molecules.[1] In this context, the substrate-controlled diastereoselective addition of organometallic reagents to aldehydes and ketones is a fundamental transformation that enables an increase in molecular complexity through C–C bond formation together with the creation of a stereogenic center. Recently, in the context of our ongoing research of the use of the boron/zinc exchange reactions for the generation of transferable aryl groups, we studied the diastereoselective arylation of chiral, nonracemic α-oxygenated aldehydes, which are readily available from carbohydrates (Scheme 1).[2] The aldehydes underwent smooth arylation with good-to-excellent diastereoselectivities, and the stereochemistry of the newly formed stereocenter was proposed to be the result of a chelation-controlled process.[3] In addition, the methodology developed has proven to be useful for the short synthesis of (+)-7-*epi*-goniofufurone, a natural product belonging to the styryllactone family and analogues thereof.[4]

In line with our previous investigation, we decided to examine whether our methodology could be applied to the arylation of chiral α-amino aldehydes. Considering the major stereochemical models, Felkin–Anh[5] and Cram-chelation,[6] usually employed to predict the stereochemical outcome of diastereoselective additions to carbonyl compounds,[7] two products might be expected from the addition of a nucleophile (Scheme 2). In this reaction, the substituents at the nitrogen atom are expected to play a key role in the stereo-determining step, which is either a result of a chelation-controlled or a direct addition *trans* to the



Scheme 1. Diastereoselective arylation of α-oxygenated aldehydes and its application in the synthesis of (+)-7-*epi*-goniofufurone and its analogues.



Scheme 2. Felkin–Anh and chelation models for the addition of a nucleophile to chiral α-amino aldehydes.

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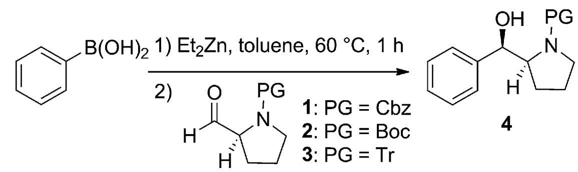
nitrogen atom, which has the lowest σ-antibonding orbital, as predicted by the Felkin–Anh model.

Importantly, the stereoselective synthesis of amino alcohols has received considerable attention over the years,[8] because they are present as structural subunits in bioactive natural products, for example, in the amino acids serine and threonine, the carbohydrate D-glucosamine, the hormones epinephrine and norepinephrine, and in a number of biologically important molecules such as ephedrine, sphingosine, nojirimycin, propanonol, and docetaxel, to cite just a few. Chiral nonracemic amino alcohols presenting two vicinal stereocenters have also been described as inhibitors of the glycine transporter enzyme GlyT1.[9] The inhibition of this enzyme leads to an increase in brain activity and might lead to agents for the treatment of diseases in which cognitive processes are related, such as schizophrenia, dementia, and Alzheimer's desease. Furthermore, amino alcohols are an important class of compounds that have been used extensively in asymmetric synthesis as building blocks, chiral auxiliaries, chiral ligands, and catalysts.[10]

Herein we report our results on the use of boron/zinc exchange reactions,[11] using arylboronic acids as the source of transferable aryl groups, for the highly diastereoselective arylation of N-protected L-prolinal.

Table 1. Screening of the reaction conditions for the phenyl transfer to N-protected L-prolinals **1**–**3**.[a]



Entry	Aldehyde	Solvent	T [°C]	t [h]	Yield [%][b]	dr[c]
1	**1** (PG = Cbz)	toluene	25	4	23	1:1
2	**2** (PG = Boc)	toluene	25	4	63	4:1
3	**3** (PG = Tr)	toluene	25	4	53	>20:1
4	**3** (PG = Tr)	toluene	25	3	46	>20:1
5	**3** (PG = Tr)	toluene	25	2	21	>20:1
6	**3** (PG = Tr)	toluene	25	4	30	>20:1
7	**3** (PG = Tr)	toluene	25	17	–	–
8[d]	**3** (PG = Tr)	toluene	25	4	76	>20:1
9	**3** (PG = Tr)	toluene/DCM	25	4	48	>20:1
10	**3** (PG = Tr)	toluene/THF	25	4	17	>20:1
11	**3** (PG = Tr)	toluene	0	8	54	>20:1
12	**3** (PG = Tr)	toluene	60	1	58	>20:1

[a] Reactions were performed with aldehyde (1.0 equiv.), $PhB(OH)_2$ (2.4 equiv.) and Et_2Zn (1.5 M solution in toluene, 7.2 equiv.). [b] Isolated yields. [c] Determined by ^1H NMR spectroscopy. Absolute configuration was assigned based on literature NMR spectroscopic data. [d] The organic layer was washed with aqueous $NaCl/NH_4Cl$ (2:1, w/w) after quenching with water.

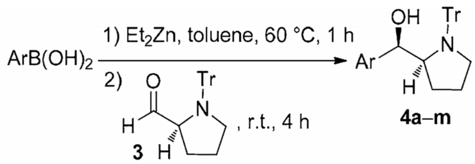
Results and Discussion

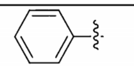
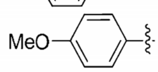
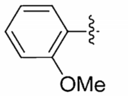
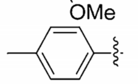
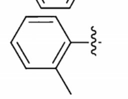
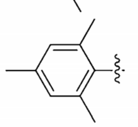
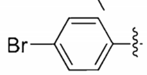
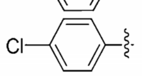
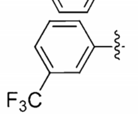
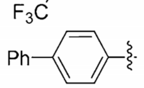
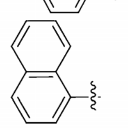
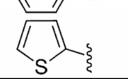
We started our investigation by examining the influence of the N-protecting group on the yield and diastereoselectivity of the reaction. Phenylboronic acid was chosen as the source of the reactive aryl group, which, by reaction with diethylzinc in toluene at 60 °C for 1 h, generates in situ the mixed PhZnEt reagent. When the Cbz-protected aldehyde[12] **1** was used, the desired product was formed in low yield and in a completely unselective reaction (Table 1, Entry 1). Changing the carbamate protecting group to the *tert*-butyloxy derivative (PG = Boc; **2**)[12] the product was obtained in 63% yield and in a moderate diastereoselectivity of 4:1 (Entry 2). On the other hand, when the protecting group at the nitrogen atom was changed to the triphenylmethyl (Tr) group, aldehyde **3**[13] underwent the phenyl transfer reaction to deliver the desired product **4** in 53% yield and with an excellent *dr* of >20:1 (Entry 3). The absolute stereochemistry of the newly generated stereocenter was assigned to (*R*) by comparison with literature NMR spectroscopic data.[13] The observed absolute configuration corresponds to the Felkin–Anh addition product and is the result of a nonchelating pathway. The same behavior was observed in the addition of Grignard reagents to N-Tr-L-prolinal[13] and in the reduction of N-Tr-α-amino ketones.[14] This result contrasts the results we had previously obtained by using α-oxygenated aldehydes, which underwent chelation-controlled addition.[2] This different reaction pathway is likely to be a result of the steric hindrance posed by the bulky trityl group on the nitrogen atom.[15]

With this highly diastereoselective reaction in hand, we screened a variety of reaction conditions in an attempt to improve the yield of the product. Reducing the reaction time resulted in a decrease in yield (Entries 4 and 5), as did increasing the reaction time. For example, if the reaction mixture was stirred at room temperature for 4 h, the yield decreased to 30% (Entry 6), and with longer reaction times (17 h; Entry 7), no product was isolated due to extensive decomposition. Because the product was prone to decomposition, we changed the workup procedure. Thus, an increase in yield was achieved when, after quenching the reaction with water and EtOAc, the organic layer was washed with an aqueous solution of $NaCl/NH_4Cl$ (2:1, w/w). Under these conditions, the product was obtained in 76% yield and with a *dr* >20:1 (Entry 8). Finally, the addition of cosolvents such as dichloromethane (Entry 9) and THF (Entry 10), or variation of the temperature from 0 (Entry 11) to 60 °C (Entry 12), did not result in any further improvement in the yield of product **4**.

Having successfully achieved phenyl transfer to aldehyde **3**, we extended our studies to a broader range of arylboronic acids as precursors for the transferable aryl groups. The influence of electron-donating and -withdrawing groups in the aryl moiety was examined. The use of arylboronic acids substituted with methoxy and methyl groups at either the *ortho* or *para* positions resulted in the corresponding amino alcohols **4b**–**4e** in moderate to good isolated yields and with excellent diastereoselectivities (Table 2, Entries 2–5). Unfortunately, the reaction with the more heavily substituted mesityl derivative failed to deliver the corresponding product **4f**, which shows that the reaction is sensitive to the presence of two *ortho* substituents at the aryl nucleophile.

Table 2. Diastereoselective arylation of *N*-Tr-L-prolinal **3**.[a]

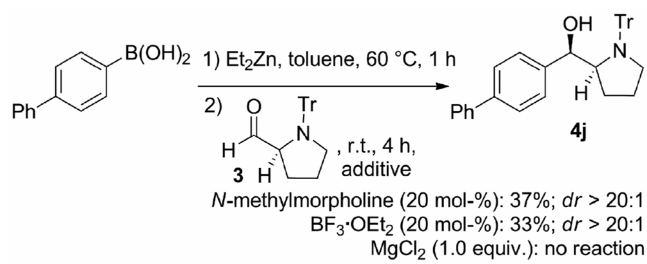


Entry	Ar	Product (yield [%])[b]	*dr*[c]
1		**4a** (76)	>20:1
2		**4b** (72)	>20:1
3		**4c** (58)	>20:1
4		**4d** (55)	>20:1
5		**4e** (67)	>20:1
6		**4f** (–)	–
7		**4g** (33)	>20:1
8		**4h** (62)	>20:1
9		**4i** (41)	>20:1
10 11[d]		**4j** (45) (56)	>20:1 >20:1
12		**4k** (48)	>20:1
13		**4l** (50)	>20:1

[a] Reactions were performed with aldehyde (1.0 equiv.), $ArB(OH)_2$ (2.4 equiv.) and Et_2Zn (1.5 M solution in toluene, 7.2 equiv.). [b] Isolated yields. [c] Determined by ^1H NMR spectroscopy. [d] Excess of the arylboronic acid (3.2 equiv.) and Et_2Zn (9.6 equiv.) were used.

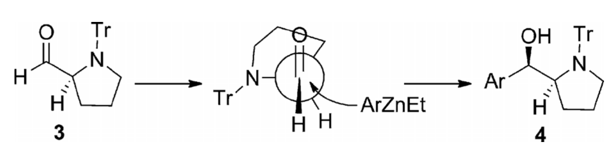
The presence of electron-withdrawing groups such as halogens (Table 2, Entries 7 and 8) or a trifluoromethyl group (Entry 9) had no impact on the diastereoselectivity of the arylation reaction, but a decrease in yield was observed. Finally, biphenyl (Entries 10 and 11), naphthyl (Entry 12), and thiophene (Entry 13) groups could also be transferred in a very stereoselective fashion, delivering the corresponding β-amino alcohols with a *dr* >20:1.

In an attempt to improve the yields of the arylation reactions, we tried to force the reaction by using a larger amount of biphenyl-4-ylboronic acid; however, only a moderate increase in yield from 45 to 56% was observed (Table 2, Entries 10 and 11). The use of additives was also tested (Scheme 3). *N*-Methylmorpholine (NMM) was added to the reaction of biphenyl-4-ylboronic acid with aldehyde **3** in an attempt to increase the reactivity of the presumed arylzinc intermediate by further coordination with a Lewis base. The presence of morpholinyl ligands in organozinc additions has been reported to be beneficial in several cases,[16] including by our group.[17] However, in this case, the use of NMM as additive resulted in a decrease in the product yield. On the other hand, the use of $BF_3{\cdot}OEt_2$ as Lewis acid to enhance the reactivity of the aldehyde also proved fruitless, delivering the product in only 33% yield. Finally, the use of $MgCl_2$ as additive has previously been reported to accelerate the addition of organozinc reagents to carbonyl compounds.[18] Unfortunately, in our system no reaction was observed in the presence of 1 equiv. of $MgCl_2$.



Scheme 3. Influence of additives on the arylation reaction.

It is important to note that, in all the reactions studied, excellent diastereoselectivity was achieved in favor of the Felkin–Anh product, regardless of the arylboronic acid used (Scheme 4). The presence of the bulky triphenylmethyl group at the nitrogen atom prevents coordination of the zinc atom, thereby prohibiting the formation of the Cram-chelation transition state. The attack of the nucleophile therefore occurs at the energetically most favorable position and *anti* to the nitrogen atom, thus avoiding steric interactions with the trityl group.



Scheme 4. Proposed transition state for the diastereoselective arylation reaction.

This proposed mechanism explains the stereochemistry observed at the new stereocenter. The lack of bidentate coordination would also account for the moderate yields of several reactions, because the chelation between the heteroatom at the α-position and the carbonyl group enhances its electrophilicity towards nucleophilic attack.

Conclusions

We have described the highly diastereoselective arylation of *N*-trityl-L-prolinal (**3**) involving the boron/zinc exchange reaction to generate the reactive aryl group. The observed stereochemistry is proposed to result from a Felkin–Anh transition state, with the competing Cram-chelation pathway prevented by the presence of the very bulky trityl group at the endocyclic nitrogen atom of the pyrrolidine ring. The resulting amino alcohols possess interesting structures that might be useful as platforms for the preparation of chiral ligands and catalysts for metal-catalyzed reactions and organocatalysis.

Experimental Section

General: NMR spectra were recorded either with a Varian 300 MHz or a Bruker 400 MHz instrument in $CDCl_3$ solutions. Chemical shifts are reported in ppm, referenced to the solvent peak of residual $CHCl_3$ or tetramethylsilane (TMS) as reference. The data are reported as follows: chemical shift (δ), multiplicity, coupling constant (J) in Hz, and integrated intensity. ^{13}C NMR spectra were recorded at 75 MHz in $CDCl_3$ solutions. Chemical shifts are reported in ppm, referenced to the solvent peak $CDCl_3$. Abbreviations to denote the multiplicity of a particular signal are: s (singlet), d (doublet), t (triplet), dd (double doublet), m (multiplet), and br. s (broad singlet). Infrared spectra (neat) were recorded with a Varian FT-IR-640 spectrometer. Optical rotations were measured with a Perkin–Elmer 341 Polarimeter. HR mass spectra (ESI) were recorded with an Orbitrap mass spectrometer (Thermo Fisher Scientific). Column chromatography was performed by using silica gel (230–400 mesh) according to the methods described by Still et al.[19] TLC was performed by using silica gel GF254 (0.25 mm thickness). For visualization, TLC plates were either placed under ultraviolet light, or treated with phosphomolybdic acid followed by heating. Air- and moisture-sensitive reactions were conducted in flame- or oven-dried glassware equipped with tightly fitted rubber septa and under a positive pressure of dry argon. Reagents and solvents were handled by using standard syringe techniques. Temperatures above room temperature were maintained by use of a mineral oil bath heated on a hotplate.

General Procedure for the Aryl Transfer Reaction: A 1.5 M solution of Et_2Zn (7.2 equiv., 3.6 mmol, 2.4 mL) was slowly added to a solution of arylboronic acid (2.4 equiv., 1.2 mmol) in dry toluene (2 mL) under argon. The mixture was stirred at 60 °C for 1 h and then cooled to room temperature. Subsequently, a solution of the aldehyde (0.5 mmol) in dry toluene (1 mL) was added. The reaction mixture was stirred at room temperature for 4 h, and then water (5 mL) was carefully added at 0 °C. The product was then extracted with ethyl acetate (3 × 10 mL) and washed with $NaCl/NH_4Cl$ (2:1, w/w) solution. The combined extracts were dried with $MgSO_4$ and the solvents evaporated. The residue was purified by flash chromatography (hexane/EtOAc/Et_3N, 9:1:0.2).

(*R*)-Phenyl[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4a): Obtained as a white solid in 76% yield (0.160 g, 0.38 mmol). M.p. 71–72 °C. $[\alpha]_D^{20}$ = –59 (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3413, 3067, 2969, 2870, 1945, 1809, 1587, 1489, 1328, 1204, 1019, 909, 736, 724 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.60 (d, J = 7.16 Hz, 6 H), 7.27–7.07 (m, 14 H), 5.12 (d, J = 3.31 Hz, 1 H), 3.74–3.69 (m, 1 H), 3.33–3.23 (m, 1 H), 3.07–2.99 (m, 1 H), 1.49–1.37 (m, 1 H), 1.35–1.25 (m, 1 H), 0.84–0.72 (m, 1 H), 0.25–0.09 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 146.9, 144.5, 142.2, 129.8, 127.9, 127.5, 127.1, 126.4, 126.3, 125.3, 78.1, 75.3, 66.1, 53.1, 25.7, 24.7 ppm. HRMS (ESI): calcd. for $[C_{30}H_{29}NO - H]^+$ 418.2176; found 418.2203.

(*R*)-(4-Methoxyphenyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4b): Obtained as a white solid in 72% yield (0.161 g, 0.36 mmol). M.p. 147 °C (dec.). $[\alpha]_D^{20}$ = –96 (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3557, 3440, 3064, 2949, 2856, 1618, 1509, 1441, 1324, 1232, 1155, 1087, 1032, 902, 840, 748, 701 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.59 (d, J = 7.03 Hz, 6 H), 7.29–7.28 (m, 3 H), 7.25–7.14 (m, 6 H), 7.00 (d, J = 8.79 Hz, 2 H), 6.78 (d, J = 8.79 Hz, 2 H), 5.07 (d, J = 3.51 Hz, 1 H), 3.73 (s, 3 H), 3.69–3.63 (m, 1 H), 3.32–3.22 (m, 1 H), 3.07–2.99 (m, 1 H), 1.49–1.38 (m, 1 H), 1.35–1.22 (m, 1 H), 0.84–0.72 (m, 1 H), 0.28–0.11 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 158.8, 145.2, 135.0, 130.5, 128.2, 127.1, 126.9, 114.0, 78.7, 75.9, 66.7, 55.8, 53.9, 26.2, 25.4 ppm. HRMS (ESI): calcd. for $[C_{31}H_{31}NO_2 - H]^+$ 448.2282; found 448.2305.

(*R*)-(2-Methoxyphenyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4c): Obtained as a white solid in 58% yield (0.130 g, 0.29 mmol). M.p. 124 °C (dec.). $[\alpha]_D^{20}$ = –71 (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3564, 3448, 3056, 2933, 2356, 1594, 1486, 1448, 1240, 1094, 1002, 894, 756, 701 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.60 (d, J = 7.04 Hz, 6 H), 7.47 (d, J = 6.46 Hz, 1 H), 7.31–7.23 (m, 11 H), 7.19–7.17 (m, 3 H), 7.14–7.11 (m, 1 H), 6.93–6.89 (t, J = 7.04 Hz, 1 H), 6.68 (d, J = 8.22 Hz, 1 H), 5.26 (d, J = 3.52 Hz, 1 H), 3.99–3.93 (m, 1 H), 3.42 (s, 3 H), 3.32–3.22 (m, 1 H), 3.13 (br. s, 1 H), 3.07–2.99 (m, 1 H), 1.47–1.36 (m, 1 H), 1.32–1.27 (m, 1 H), 0.76–0.64 (m, 1 H), 0.20–0.07 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 146.8, 144.6, 130.4, 129.8, 127.9, 127.4, 127.2, 126.4, 126.0, 120.3, 78.1, 71.6, 62.7, 54.9, 53.3, 26.0, 24.9 ppm. HRMS (ESI): calcd. for $[C_{31}H_{31}NO_2 - H]^+$ 448.2282; found 448.2311.

(*R*)-(*p*-Tolyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4d): Obtained as a white solid in 55% yield (0.120 g, 0.28 mmol). M.p. 116 °C (dec.). $[\alpha]_D^{20}$ = –76 (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3472, 3064, 3018, 2971, 2849, 1602, 1486, 1441, 1324, 1187, 1147, 1079, 1010, 894, 840, 756, 694 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.60 (d, J = 7.03 Hz, 6 H), 7.24–7.12 (m, 9 H), 7.03 (d, J = 8.20 Hz, 2 H), 6.96 (d, J = 8.20 Hz, 2 H), 5.01 (d, J = 2.93 Hz, 1 H), 3.74–3.65 (m, 1 H), 3.32–3.23 (m, 1 H), 3.06–2.98 (m, 1 H), 2.25 (s, 3 H), 1.50–1.37 (m, 1 H), 1.35–1.28 (m, 1 H), 0.87–0.72 (m, 1 H), 0,26–0.01 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 144.5, 139.1, 135.8, 129.8, 128.5, 27.8, 127.8, 127.4, 127.1, 126.2, 125.2, 78.0, 75.5, 66.0, 53.1, 25.5, 24.7, 21.0 ppm. HRMS (ESI): calcd. for $[C_{31}H_{31}NO - H]^+$ 432.2333; found 432.2358.

(*R*)-(*o*-Tolyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4e): Obtained as a white solid in 67% yield (0.144 g, 0.33 mmol). M.p. 60–62 °C. $[\alpha]_D^{20}$ = –115 (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3433, 3056, 2956, 1594, 1486, 1433, 1309, 1194, 1102, 1010, 910, 833, 763, 709 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.59 (d, J = 7.04 Hz, 6 H), 7.54 (m, 1 H), 7.29–7.24 (m, 6 H), 7.21–7.11 (m, 4 H), 7.08–7.03 (m, 1 H), 6.97–6.94 (m, 1 H), 5.21 (d, J = 3.52 Hz, 1 H), 3.85–3.78 (m, 1 H), 3.30–3.21 (m, 1 H), 3.09–3.02 (m, 1 H), 1.60 (s, 3 H), 1.54–1.42 (m, 1 H), 1.31–1.26 (m, 1 H), 0.76–0.64 (m, 1 H), 1.07 to –0.02 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 147.0, 144.3, 139.8, 133.8, 130.0, 129.9, 128.0, 127.9, 127.7, 127.3, 126.5, 126.4, 125.9, 125.6, 78.2, 72.5, 62.6, 53.6, 26.2, 24.8, 18.4 ppm. HRMS (ESI): calcd. for $[C_{31}H_{31}NO - H]^+$ 432.2333; found 432.2370.

(*R*)-(4-Bromophenyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4g): Obtained as a white solid in 33% yield (0.082 g, 0.16 mmol). M.p. 117 °C (dec.). $[\alpha]_D^{20} = -39$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3479, 3048, 2933, 1586, 1501, 1448, 1324, 1155, 1002, 871, 756, 686 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.57 (d, J = 8.22 Hz, 6 H), 7.24–7.15 (m, 9 H), 6.94 (d, J = 8.22 Hz, 2 H), 5.01 (d, J = 3.52 Hz, 1 H), 3.70–3.64 (m, 1 H), 3.29–3.20 (m, 1 H), 3.06–2.99 (m, 1 H), 2.89 (br. s, 1 H), 1.41–1.28 (m, 2 H), 0.82–0.73 (m, 1 H), 0.24–0.15 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 147.5, 145.0, 141.7, 131.7, 130.5, 128.5, 128.2, 127.9, 127.8, 127.0, 120.8, 78.7, 75.5, 66.3, 53.9, 26.5, 25.3 ppm. HRMS (ESI): calcd. for $[C_{30}H_{28}BrNO - H]^+$ 496.1282; found 496.1295.

(*R*)-(4-Chlorophenyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4h): Obtained as a white solid in 62% yield (0.141 g, 0.31 mmol). M.p. 125–128 °C. $[\alpha]_D^{20} = -68$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3464, 3064, 2956, 2849, 1602, 1486, 1433, 1309, 1163, 1079, 1017, 902, 848, 778, 701 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.58 (d, J = 7.45 Hz, 6 H), 729–723 (m, 9 H), 7.19 (m, 2 H), 6.99 (d, J = 8.31 Hz, 2 H), 5.03 (d, J = 3.50 Hz, 1 H), 3.71–3.65 (m, 1 H), 3.30–3.21 (m, 1 H), 3.07–3.00 (m, 1 H), 2.82 (br. s, 1 H), 1.42–1.3 (m, 1 H), 0.88–0.73 (m, 1 H), 0.27–0.10 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 144.4, 129.9, 128.1, 127.9, 127.6, 127.3, 126.8, 126.4, 82.0, 74.9, 65.8, 53.4, 25.9, 24.7 ppm. HRMS (ESI): calcd. for $[C_{30}H_{28}ClNO - H]^+$ 452.1787; found 452.1795.

(*R*)-[3-(Trifluoromethyl)phenyl][(*S*)-1-tritylpyrrolidin-2-yl]methanol (4i): Obtained as a white solid in 41% yield (0.099 g, 0.20 mmol). M.p. 107 °C (dec.). $[\alpha]_D^{20} = -13$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3625, 3464, 3056, 2949, 2856, 1971, 1733, 1586, 1486, 1433, 1324, 1155, 1117, 1010, 902, 756, 694, 624 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.58 (d, J = 7.04 Hz, 6 H), 7.43–7.39 (m, 1 H), 7.25–7.16 (m, 13 H) 5.06 (d, J = 2.93 Hz, 1 H), 3.75–3.69 (m, 1 H), 3.30–3.21 (m, 1 H), 3.08–3.00 (m, 1 H), 1.42–1.28 (m, 2 H), 0.89–0.74 (m, 1 H), 0.29–0.13 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 146.9, 144.4, 143.2, 128.0, 127.7, 127.3, 126.5, 123.4, 122.4, 78.1, 74.9, 65.7, 53.4, 26.0, 24.7 ppm. HRMS (ESI): calcd. for $[C_{31}H_{28}F_3NO - H]^+$ 486.2050; found 486.2069.

(*R*)-(Biphenyl-4-yl)[(S)-1-tritylpyrrolidin-2-yl]methanol (4j): Obtained as a white solid in 56% yield (0.138 g, 0.28 mmol). M.p. 67 °C (dec.). $[\alpha]_D^{20} = -73$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3456, 3064, 2925, 1578, 1486, 1441, 1317, 1155, 1002, 763, 694 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.61 (d, J = 7.62 Hz, 6 H), 7.54 (d, J = 7.62 Hz, 2 H), 7.47 (d, J = 8.20 Hz, 2 H), 7.40 (m, 2 H), 7.32–7.26 (m, 9 H), 7.23–7.14 (m, 3 H), 5.14 (d, J = 2.93 Hz, 1 H), 3.78–3.73 (m, 1 H), 3.35–3.25 (m, 1 H), 3.09–3.01 (m, 1 H), 1.53–1.42 (m, 1 H), 1.37–1.29 (m, 1 H), 0.89–0.79 (m, 1 H), 0.29–0.14 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.9, 144.5, 141.2, 140.9, 139.3, 129.8, 128.6, 127.9, 127.5, 127.4, 127.2, 127.0, 126.9, 126.6, 126.3, 125.8, 78.1, 75.4, 65.9, 53.3, 25.8, 24.7 ppm. HRMS (ESI): calcd. for $[C_{36}H_{33}NO - H]^+$ 494.2489; found 494.2498.

(*R*)-(1-Naphthyl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4k): Obtained as a white solid in 48% yield (0.112 g, 0.24 mmol). M.p. 126 °C (dec.). $[\alpha]_D^{20} = -110$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3564, 3071, 2941, 1586, 1479, 1441, 1217, 1087, 1017, 910, 801, 748, 686 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.77 (d, J = 8.22 Hz, 1 H), 7.70 (d, J = 8.22 HZ, 1 H), 7.64 (d, J = 7.04 Hz, 6 H), 7.45–7.34 (m, 5 H), 7.27–7.21 (m, 6 H), 7.11–7.05 (m, 1 H), 6.41 (d, J = 8.22 Hz, 1 H), 5.81 (d, J = 2.93 Hz, 1 H), 4.10–4.00 (m, 1 H), 3.34–3.24 (m, 1 H), 3.13–3.05 (m, 1 H), 1.59–1.47 (m, 1 H), 1.34–1.25 (m, 1 H), 0.68–0.57 (m, 1 H), –0.03 to –0.19 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 164.9, 146.8, 144.4, 137.1, 133.3, 130.1, 128.6, 127.9, 127.7, 127.2, 127.0, 126.4, 125.5, 125.3, 125.0, 122.8, 122.5, 78.1, 72.8, 64.2, 53.8, 25.8, 24.8 ppm. HRMS: calcd. for $[C_{34}H_{31}NO - H]^+$ 468.2333; found 468.2409.

(*S*)-(Thiophen-2-yl)[(*S*)-1-tritylpyrrolidin-2-yl]methanol (4l): Obtained as a thick oil in 50% yield (0.106 g, 0.25 mmol). $[\alpha]_D^{20} = -45$ (c = 1.0, EtOAc). IR: $\tilde{\nu}$ = 3456, 2910, 2348, 1578, 1479, 1441, 1017, 902, 740, 686 cm^{-1}. ^1H NMR ($CDCl_3$, 300 MHz): δ = 7.58 (d, J = 7.03 Hz, 2 H), 7.29–7.27 (m, 4 H), 7.24–7.16 (m, 6 H), 7.12 (dd, J^1 = 5.27, J^2 = 1.17 Hz, 1 H), 6.89–6.86 (m, 1 H), 6.66–6.65 (m, 1 H), 5.32 (d, J = 2.34 Hz, 1 H), 3.74–3.69 (m, 1 H), 3.33–3.23 (m, 1 H), 3.06–2.98 (m, 1 H), 1.65–1.54 (m, 1 H), 1.42–1.30 (m, 1 H), 1.05–0.93 (m, 1 H), 0.35–0.20 (m, 1 H) ppm. ^{13}C NMR ($CDCl_3$, 75 MHz): δ = 146.2, 144.5, 129.7, 128.0, 127.5, 127.2, 126.5, 126.3, 123.4, 122.4, 78.0, 73.8, 66.2, 53.0, 25.9, 24.8 ppm. HRMS: calcd. for $[C_{28}H_{27}NOS - H]^+$ 424.1741; found 424.1745.

Supporting Information (see footnote on the first page of this article): ^1H and ^{13} NMR spectra of synthesized compounds.

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